STATS I · WEEK 4

Stats I week 4 formulas: Association between two variables

Two categorical variables · Scatter plots & covariance · Correlation coefficient (r) · Point bi-serial correlation. Save this PDF for revision.

TWO CATEGORICAL VARIABLES · IN EVERY PAPER

Two categorical variables

Joint relative frequency

cell/(grand total)

Row relative frequency

cell/(row total)

Column relative frequency

cell/(column total)

Marginal totals

row totals and column totals; each set adds to the grand total

TWO CATEGORICAL VARIABLES · IN EVERY PAPER

Key points

- Contingency (two-way) table: one variable on the rows, the other on the columns, counts in the cells, totals in the margins.
- Association: compare row percentages across the rows. If they are (nearly) the same in every row, there is no association.
- A stacked (or 100% stacked) bar chart shows those row percentages visually.

TWO CATEGORICAL VARIABLES · IN EVERY PAPER

Common traps

⚠ Tip: Draw the empty table with its totals row and column first, write in every number given, then fill the blanks so each row and column adds up.

SCATTER PLOTS & COVARIANCE · IN EVERY PAPER

Scatter plots & covariance

Sample covariance

$$(\Sigma(x_i - \bar{x})(y_i - \bar{y}))/(n - 1) = (\Sigma\, x_i y_i - n\bar{x}\bar{y})/(n - 1)$$

Population covariance

$$(\Sigma(x_i - \mu_x)(y_i - \mu_(y)))/n$$

Scaling

$$\mathrm{cov}(ax + b, cy + d) = ac \cdot \mathrm{cov}(x, y)$$

With itself

$$\mathrm{cov}(x, x) = \mathrm{Var}(x)$$

SCATTER PLOTS & COVARIANCE · IN EVERY PAPER

Key points

- Sign gives direction: positive (rise together), negative (one rises as the other falls), near 0 (no linear relation).
- Size depends on the units, so covariance can't measure strength. That's what correlation is for.
- Reading a scatter plot: direction, form (straight or curved), strength (how tightly points hug a line), outliers.

SCATTER PLOTS & COVARIANCE · IN EVERY PAPER

Common traps

⚠ PYQs ask both kinds: 'population covariance' ⇒ ÷ n, 'sample covariance' ⇒ ÷ (n – 1).

CORRELATION COEFFICIENT (R) · IN EVERY PAPER

Correlation coefficient (r)

Pearson r

$$r = (cov(x, y))/(s_x\ s_(y))$$

Use sample values throughout, or population values throughout. Never mix.

Computational form

$$r = (n\Sigma xy - \Sigma x\Sigma y)/(\sqrt{([n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}))$$

Range

$$-1 \le r \le 1$$

Change of origin and scale

$$r(ax + b, cy + d) = r(x, y) \text{ if } ac > 0, -r(x, y) \text{ if } ac < 0$$

CORRELATION COEFFICIENT (R) · IN EVERY PAPER

Reading r

r	Meaning
+1 or –1	Perfect straight line, rising or falling
Close to ±1	Strong linear association
Close to 0	Weak or no linear association (could still be curved)
Sign	Direction only: + rising, – falling

CORRELATION COEFFICIENT (R) · IN EVERY PAPER

Common traps

⚠ r measures how tightly the points hug a line, not how steep it is. A steep loose cloud has a smaller |r| than a shallow tight one.

⚠ Ordering scatter plots by r: strong negative < weak negative < none < weak positive < strong positive.

⚠ r = 0 doesn't mean no relationship: a U-shape can have r ≈ 0.

⚠ r has no units, so changing units (cm → m, ₹ → $) never changes it.

⚠ Correlation is not causation.

POINT BI-SERIAL CORRELATION · OFTEN ASKED

Point bi-serial correlation

When

one variable is dichotomous (0/1: pass/fail, male/female), the other numerical

With sample SD

$$r_{pb} = (\bar{Y}_1 - \bar{Y}_0)/s_Y \times \sqrt{(n_1 n_0/(n(n - 1)))}$$

With population SD

$$r_{pb} = (\bar{Y}_1 - \bar{Y}_0)/\sigma_Y \times \sqrt{(n_1 n_0/n^2)}$$

Same answer as the sample version.

Symbols

$\bar{Y}_1$, $\bar{Y}_0$: group means · n_1, n_0: group sizes · $n = n_1 + n_0$ · s_Y, σ_Y: SD of all Y values

POINT BI-SERIAL CORRELATION · OFTEN ASKED

Common traps

⚠ Pair the SD with the right root: sample SD ↔ $n(n-1)$; population SD ↔ n^2. Mixing them gives a wrong value.

⚠ Swapping which group is coded 1 flips the sign, which is why PYQs ask for the absolute value.

Notes for every week of Stats I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/stats-1 →

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